

An In-Depth Discussion of the Mathematical Definitions of the Vibrational Density of States

There are two ways of computing the Vibrational Density of States (VDOS): one is using the velocity power spectral density (VPSD), the other is by using the velocity autocorrelation function (VACF).

Velocity Power Spectral Density¹

The VPSD is given by²:

$$S_v(\nu) = \lim_{\tau \rightarrow \infty} \frac{1}{2\tau} \mathbb{E} \left| \int_{-\tau}^{\tau} v(t) \exp(-i2\pi\nu t) dt \right|^2 \quad 1$$

where $S_v(\nu)$ is the VPSD, τ is the finite-time window, $\mathbb{E}$ is the expected value, $v(t)$ is the velocity, ν is the frequency, and

$$|X|^2 = XX^*, \forall X = a + ib \ (a, b \in \mathbb{R}) \quad 2$$

for any signal X with complex conjugate X^* . In other words, the VPSD can also be written as follows:

$$S_v(\nu) = \lim_{\tau \rightarrow \infty} \frac{1}{2\tau} \mathbb{E} \left(\int_{-\tau}^{\tau} v(t) \exp(-i2\pi\nu t) dt \right) \left(\int_{-\tau}^{\tau} v(t) \exp(-i2\pi\nu t) dt \right)^* \quad 3$$

The expected value, $\mathbb{E}$, is necessary because it is assumed that $v(t)$ is stochastic and it would not appear in the case of purely deterministic velocities.

To further clarify, $|X|^2$ can be found by taking the magnitude squared of the signal X :

$$X = a + ib, X^* = a - ib \quad 4$$

$$XX^* = a^2 + b^2 \quad 5$$

$$|X| = \sqrt{a^2 + b^2} \quad 6$$

$$|X|^2 = a^2 + b^2 \quad 7$$

$$|X|^2 = XX^* \quad 8$$

Discretely, the VPSD is found as follows:

¹ https://en.wikipedia.org/wiki/Spectral_density

² <https://doi.org/10.1063/1.1624057>

$$S_v(v) = \frac{1}{N_t} \mathbb{E} |\mathcal{F}[v(t)](v)|^2 \quad 9$$

where N_t is the number of timesteps in the finite time window, and $\mathcal{F}[\cdot]$ is the discrete Fourier transform operator, in this case the FFT. The computational cost of the PSD scales as $\mathcal{O}(N_t \log(N_t))$, as the computational bottleneck is the Fourier transform – and FFT's cost scales as $\mathcal{O}(N_t \log(N_t))$.

Notice that until now, the VPSD definition has remained independent of the VACF. The general definition of the VPSD is independent of the VACF, they will only be related to each other through the Wiener-Khinchin theorem (not a definition, but a proved theorem) as shown further down.

Velocity Autocorrelation Function³

The VACF is given as follows:

$$C_{vv}(t_1, t_2) = \langle v(t_1)v(t_2)^* \rangle \quad 10$$

where $C_{vv}(t)$ is the VACF, $\langle \cdot \rangle$ denotes the ensemble average, $v(t)$ is the velocity at time t , and t_1, t_2 are times. This is the most general definition of the VACF, and it does not assume wide-sense stationarity (on account of this, $C_{vv}(t_1, t_2)$ is a multivariable function).

In the case of wide-sense stationarity⁴ (i.e., $\langle v(t) \rangle = \mu$ (*constant*), $\langle v(t_0)v(t_0 - t)^* \rangle = C_{vv}(t)$), the VACF is only a function of the time lag t from a reference time t_0 becoming:

$$C_{vv}(t) = \langle v(t_0)v(t_0 - t)^* \rangle \quad 11$$

The computational cost of the VACF is $\mathcal{O}(N_t^2)$ since there are N_t timesteps and $(N_t - 1)$ time lags.

Wiener-Khinchin Theorem⁵

The Wiener-Khinchin theorem states that for a wide-sense stationary process $X(t)$:

$$S_X(v) = \int_{-\infty}^{\infty} C_{XX}(t) \exp(-i2\pi vt) dt \quad 12$$

Since this is a theorem, the relation between $S_X(v)$ and $C_{XX}(t)$ cannot be a matter of definition, but rather a provable mathematical consequence in light of certain conditions (i.e., wide-sense stationarity).

³ <https://en.wikipedia.org/wiki/Autocorrelation>

⁴ https://en.wikipedia.org/wiki/Stationary_process

⁵ https://en.wikipedia.org/wiki/Wiener-Khinchin_theorem

The proof of the theorem goes as follows. Let $X(t)$ be a wide-sense stationary process. The power spectral density (PSD) is given by (1) as:

$$S_X(\nu) = \lim_{\tau \rightarrow \infty} \frac{1}{2\tau} \mathbb{E} \left| \int_{-\tau}^{\tau} X(t) \exp(-i2\pi\nu t) dt \right|^2 \quad 13$$

Which can be expanded according to (2) as:

$$S_X(\nu) = \lim_{\tau \rightarrow \infty} \frac{1}{2\tau} \mathbb{E} \left(\int_{-\tau}^{\tau} X(t) \exp(-i2\pi\nu t) dt \right) \left(\int_{-\tau}^{\tau} X(t) \exp(-i2\pi\nu t) dt \right)^* \quad 14$$

Now, the complex conjugate in the second integral can be brought inside the integral due to the linearity of integrals:

$$\left(\int_{-\tau}^{\tau} X(t) \exp(-i2\pi\nu t) dt \right)^* = \int_{-\tau}^{\tau} (X(t) \exp(-i2\pi\nu t))^* dt \quad 15$$

and then, since for any $A, B \in \mathbb{C}$:

$$(AB)^* = A^* B^* \quad 16$$

$$\left(\int_{-\tau}^{\tau} X(t) \exp(-i2\pi\nu t) dt \right)^* = \int_{-\tau}^{\tau} X(t)^* \exp(-i2\pi\nu t)^* dt \quad 17$$

simplifying the complex conjugate of the exponential in the integrand:

$$\left(\int_{-\tau}^{\tau} X(t) \exp(-i2\pi\nu t) dt \right)^* = \int_{-\tau}^{\tau} X(t)^* \exp(i2\pi\nu t) dt \quad 18$$

So using (18) expands (14) as:

$$S_X(\nu) = \lim_{\tau \rightarrow \infty} \frac{1}{2\tau} \mathbb{E} \left(\int_{-\tau}^{\tau} X(t) \exp(-i2\pi\nu t) dt \right) \left(\int_{-\tau}^{\tau} X(t)^* \exp(i2\pi\nu t) dt \right) \quad 19$$

Without loss of generality, we can change t for s in the second integral for simplification (this will make the next step significantly clearer). Fubini's theorem⁶ simplifies the product of two integrals as a double integral:

$$\int_a^b v(x) dx \int_a^b w(y) dy = \int_a^b \int_a^b v(x) w(y) dx dy \quad 20$$

Which turns (19) into:

⁶ https://en.wikipedia.org/wiki/Fubini%27s_theorem

$$S_X(\nu) = \lim_{\tau \rightarrow \infty} \frac{1}{2\tau} \mathbb{E} \int_{-\tau}^{\tau} \int_{-\tau}^{\tau} X(t) \exp(-i2\pi\nu t) X(s)^* \exp(i2\pi\nu s) dt ds \quad 21$$

Simplifying (21):

$$S_X(\nu) = \lim_{\tau \rightarrow \infty} \frac{1}{2\tau} \mathbb{E} \int_{-\tau}^{\tau} \int_{-\tau}^{\tau} X(t) X(s)^* \exp(-i2\pi\nu(t-s)) dt ds \quad 22$$

Changing variables $s = t_0 - t$:

$$S_X(\nu) = \lim_{\tau \rightarrow \infty} \frac{1}{2\tau} \mathbb{E} \int_{-\tau}^{\tau} \int_{t_0-\tau}^{t_0+\tau} X(t_0) X(t_0 - t)^* \exp(-i2\pi\nu t) dt dt_0 \quad 23$$

Since integrals are linear and the expected value does not affect the exponential term (it's deterministic):

$$S_X(\nu) = \lim_{\tau \rightarrow \infty} \frac{1}{2\tau} \int_{-\tau}^{\tau} \int_{t_0-\tau}^{t_0+\tau} \langle X(t_0) X(t_0 - t)^* \rangle \exp(-i2\pi\nu t) dt dt_0 \quad 24$$

Due to wide-sense stationarity, $\langle X(t_0) X(t_0 - t)^* \rangle$ is independent of t_0 , so:

$$S_X(\nu) = \int_{-\infty}^{\infty} \langle X(t_0) X(t_0 - t)^* \rangle \exp(-i2\pi\nu t) dt \quad 25$$

This concludes the prove that under wide-sense stationarity, Wiener-Khinchin holds and the VPSD equals the Fourier transform of the VACF.

Which One Is *THE* VDOS?

The historical development of the VDOS is very difficult to track because people have done all sorts of things (including not using the correct scaling, not including all the degrees of freedom, referencing solid-state physics books that talk about the DOS and thinking that is the VDOS), so simply following references does not converge to an origin. From my long query, the VDOS originates in trying to come up with an analogous version of the DOS in solid-state physics for fluids. Initially, the attempt was made by looking at Instantaneous Normal Modes (INMs). Somehow from there, the VDOS began being used as the mass-weighted VPSD with a normalization that ensures the fundamental properties of the VDOS are maintained (i.e., the integral over all real positive frequencies gives the number of degrees of freedom). This is further complicated by the fact that even in statistics textbooks, Wiener-Khinchin theorem is implied and it's often stated that by definition:

$$S_X(\nu) = \int_{-\infty}^{\infty} X(t) \exp(-i2\pi\nu t) dt \quad 26$$

which clearly cannot be the definition because otherwise Wiener-Khinchin would not be a theorem. So it is unclear whether one of the definitions – the one from the VPSD or the one from

the VACF is the *correct* one – however, it does not seem to matter because it is necessarily the case that the velocities be wide-sense stationary for the adequate use of the equipartition theorem – which ties the integral of the VDOS over all real positive frequencies to the number of degrees of freedom. Since both – the VPSD and the VACF – sides are mathematically equal under wide-sense stationarity, the argument of which is the *correct* one is meaningless.

Mathematical Definition of the VDOS

From the above discussion, the two mathematical definitions of the VDOS are:

$$g(\nu) = \frac{2}{k_B T} \sum_{i=1}^N \sum_{\alpha=1}^d m_i S_{\nu}(\nu) \quad 27$$

$$g(\nu) = \frac{2}{k_B T} \sum_{i=1}^N \sum_{\alpha=1}^d m_i \int_{-\infty}^{\infty} C_{\nu\nu}(t) \exp(-i2\pi\nu t) dt \quad 28$$

These two definitions are equivalent in the case of wide-sense stationary velocities, which is the case for equilibrium systems. Since equilibrium is required for equipartition, and equipartition is necessary for the fundamental property of the VDOS – that the integral over all real positive frequencies equals the number of degrees of freedom – to be fulfilled, neither seems to be more *correct* or *fundamental* than the other.